



SPUTTERONE
PHYSICS-INFORMED SIMULATION

A Physics-Informed Interactive Magnetron Sputtering Simulator
Enabling a Process Digital Twin

USER MANUAL

Version 1.0.1

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1 - Introduction and Overview

SputterOne is a browser-based, physics-informed interactive simulation platform for real-time exploration of magnetron sputtering process conditions. It integrates Yamamura-Bohdansky sputtering yield formulation, kinetic-theory mean free path transport, Berg-framework reactive sputtering hysteresis, co-sputtering composition estimation, and Bauer-scheme thin-film growth mode classification within a unified computational environment.

SputterOne is designed for researchers familiar with magnetron sputtering who want to explore process parameter dependencies interactively before conducting physical experiments. The desktop application (Windows) is the recommended installation for regular use and provides full offline functionality. Registration is required before download - visit the Optoelectronics Device Laboratory website to register and obtain the installer.

Feature	Description
9 Target Materials	Cu, Al, Ti, Zn, Sn, Ag, W, Ta, Mo with Yamamura-Bohdansky parameters
3 Substrate Options	Si(100), Glass (SiO ₂), Sapphire (Al ₂ O ₃) with Bauer delta-gamma criterion
Multi-Gun Co-Sputtering	Up to 3 simultaneous independent magnetron guns
Reactive Sputtering	Ar/O ₂ with Berg-framework poisoning and hysteresis
Process Advisory	Physics-informed parameter pre-screening before simulation
CSV Export	Time series, radial thickness profile, process summary
Tutorial + Labs	14-chapter tutorial and 5 structured virtual laboratory modules

Important note: *SputterOne is a reduced-order model validated for qualitative trend pre-screening. It correctly classifies transport regimes, yield trends, and growth modes, but does not provide absolute deposition rate predictions calibrated to specific systems. Results should be interpreted as physical guidance, not process recipes.*

2 - Getting Started

2.1 Desktop Application (Windows) - Recommended

The standalone Windows desktop application (SputterOne-v1.0.1-Setup.exe) is the recommended installation for all users. It runs fully offline without a browser or internet connection and provides the best performance.

To download the installer:

1. Visit the Optoelectronics Device Laboratory website:
balaji-gururajan90.github.io/Optoelectronics-device-laboratory-/sputterone.html
2. Complete the registration form (Name, Email, Institution, Purpose).
3. Agree to cite SputterOne if used in published work.
4. Click "Download SputterOne v1.0.1 (.exe)" - the installer downloads automatically.
5. Run SputterOne-v1.0.1-Setup.exe and follow the installation wizard.
6. Launch SputterOne from the Start menu or desktop shortcut.

***Note:** If Windows Defender SmartScreen shows a security warning, click "More info" then "Run anyway." SputterOne is fully open source - the complete source code is available at github.com/balaji-gururajan90/SputterOne for independent verification.*

2.2 System Requirements

Component	Minimum	Recommended
OS	Windows 10 64-bit	Windows 11 64-bit
RAM	4 GB	8 GB
Display	1280x720	1920x1080 or higher
Storage	200 MB free	500 MB free
Internet	Required for registration/download only	Not required after installation
macOS	Not yet supported (desktop app)	Use browser version at github.io/SputterOne

3 - Interface Overview

The SputterOne interface is divided into four main areas.

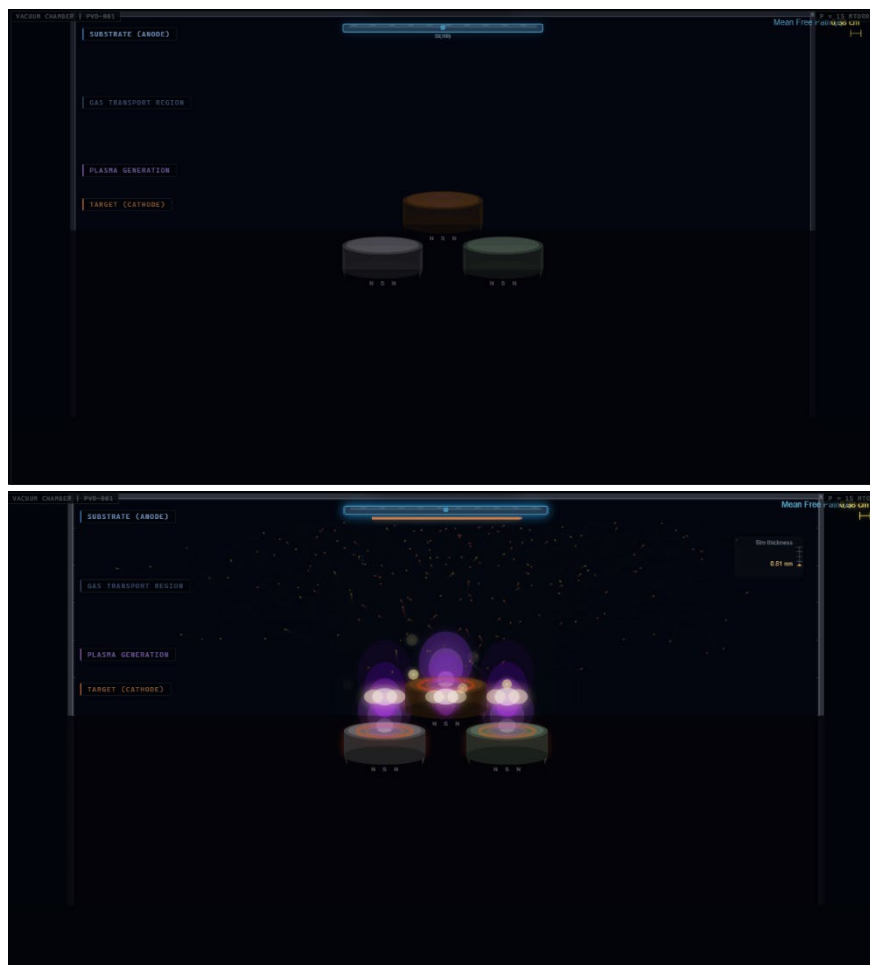
3.1 Navigation Bar

The fixed top navigation bar provides access to the main simulator, tutorial, laboratory modules, physics reference, and about page. The active section is highlighted. The SputterOne logo on the left returns to the home page.

3.2 Chamber Canvas (Centre)

The chamber canvas occupies the central area and shows real-time particle physics visualisation. It displays:

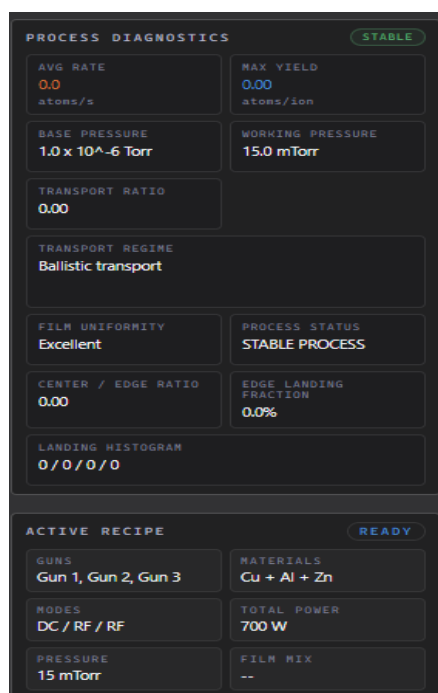
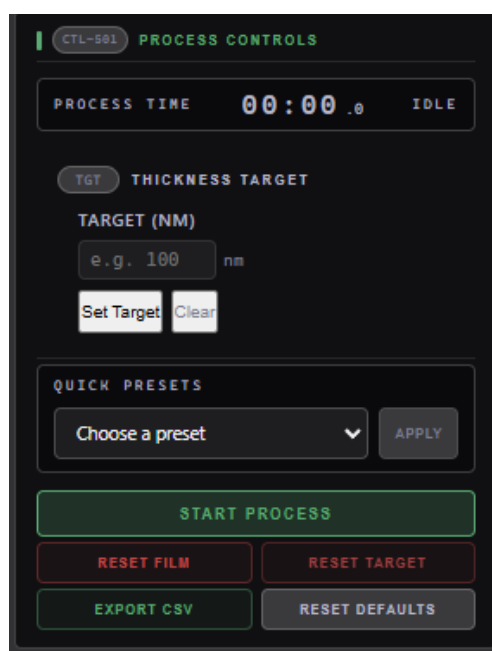
- Target (cathode) at the bottom with magnetron magnet configuration (N-S-N poles)
- Substrate (anode) at the top
- Plasma generation region showing $E \times B$ confinement zone
- Gas transport region labelled with current mean free path
- Sputtered atom particles travelling from target to substrate in real time
- Working pressure displayed in top-right corner
- Film thickness indicator at substrate edge



3 - Interface Overview

3.3 Right Sidebar Panel

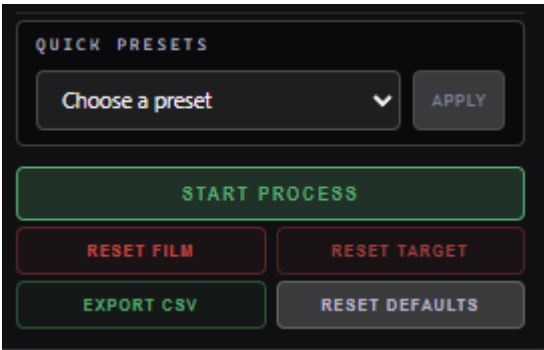
Panel	Contents
Process Controls	Process timer, thickness target, Set/Clear buttons
Process Diagnostics	All live physics outputs - yield, MFP, transport regime, deposition rate, uniformity, growth mode, delta-gamma, substrate
Active Recipe	Summary of current gun configurations and total power
Charts	Time series chart and radial thickness profile - switchable via tabs
Intelligent Recipe Advisor	Collapsible - click header to expand. Physics-informed parameter assessment with recommendations



3.4 Bottom Control Strip

Button	Function
Start Process / Pause	Starts or pauses simulation. Button turns amber when paused.
Reset Film	Clears accumulated film thickness and resets deposition charts.
Reset Target	Resets target erosion state and poisoning fraction.
Export CSV	Downloads simulation data as CSV file (time series, radial profile, summary).
Reset Defaults	Resets all parameters to default values.

4 - Setting Process Parameters

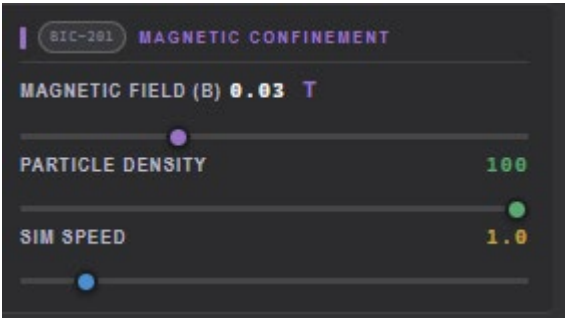
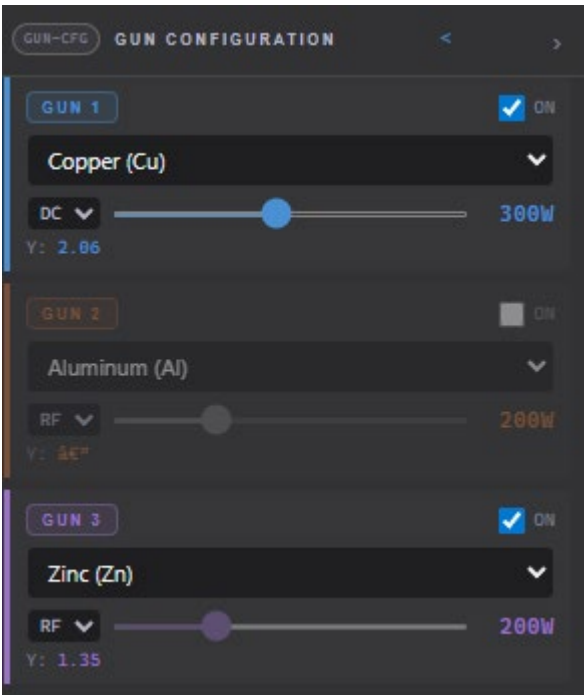


All process parameters are set in the left sidebar panel. Parameters are organised into collapsible sections. Click any section header to expand or collapse it.

4.1 Gun Configuration

SputterOne supports up to three simultaneous magnetron guns. By default only Gun 1 is active. Enable additional guns using the toggle switch next to each gun header.

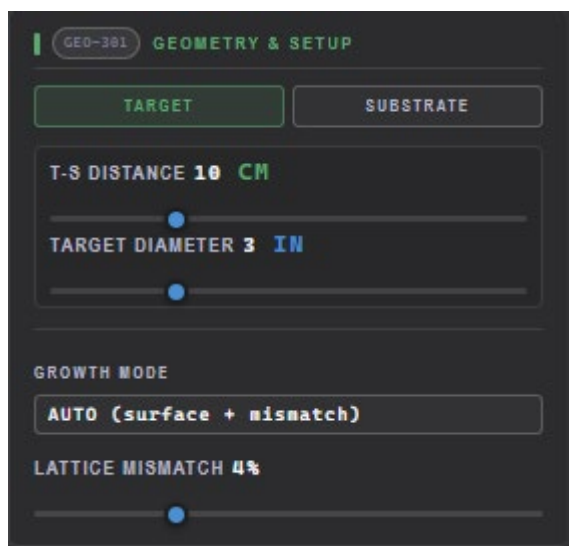
Parameter	Range	Description
Target Material	Cu Al Ti Zn Sn Ag W Ta Mo	Selects target material. Updates Yamamura-Bohdansky parameters automatically.
Operating Mode	DC / RF	DC gives higher per-ion energy. RF gives lower ion energy ($f_{RF} = 0.60$) but higher ionisation efficiency. Use DC for metallic films, RF for reactive deposition.
Power (W)	50-500 W	Applied forward power. Higher power increases discharge voltage and ion energy, increasing sputtering yield.
Magnetic Field	0.04-0.20 T	Relative magnetic confinement strength. Stronger field increases $E \times B$ electron confinement and ionisation efficiency.



4 - Setting Process Parameters

4.2 Geometry and Setup

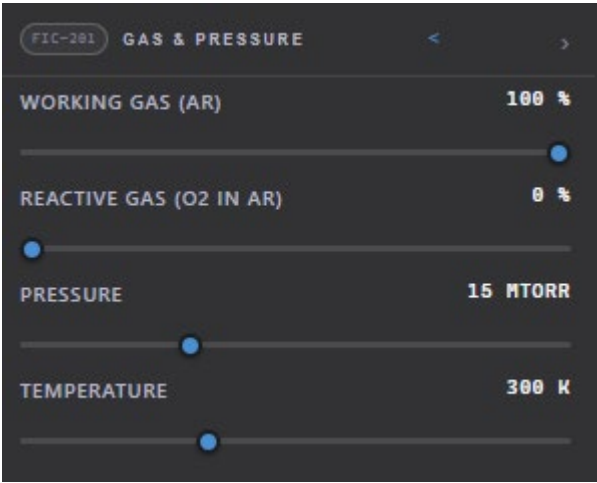
Parameter	Range	Description
Target Diameter	1-4 inch	Diameter of sputtering target. Affects spatial flux distribution and coverage ratio.
Substrate Diameter	2-12 inch	Diameter of substrate. Used to compute film uniformity NU% and coverage ratio.
Target-Substrate Distance	3-20 cm	Distance d from target to substrate. Scattering ratio $\chi = d/\lambda$ computed from this value.
Substrate Rotation	On/Off	Enables continuous substrate rotation for improved uniformity.
Lattice Mismatch	0-15%	User-defined lattice mismatch. Values above 7% favour SK growth mode.
Substrate Temperature	250-600 K	Affects surface adatom mobility influencing growth mode.



4.3 Gas Composition

Parameter	Range	Description
Working Pressure	0.1-30 mTorr	Ar working gas pressure. Controls mean free path and scattering ratio. Discharge stability warning below 0.5 mTorr.
O2 Fraction	0-50%	Oxygen fraction for reactive sputtering. Berg-framework model activates when O2 > 0.

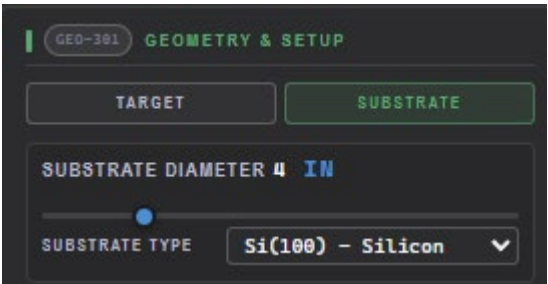
4 - Setting Process Parameters



4.4 Substrate Selection

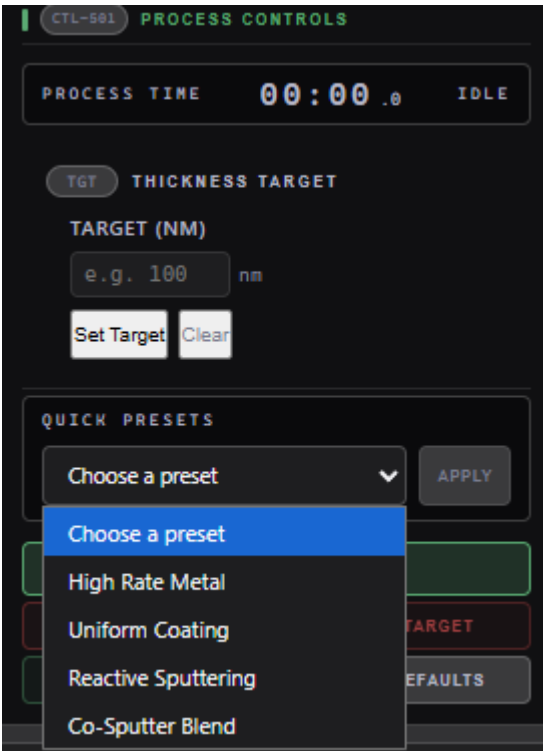
Select substrate material from dropdown in Geometry panel. Determines γ_{sub} in Bauer criterion $\Delta\gamma = \gamma_{film} + \gamma_{int} - \gamma_{sub}$.

Substrate	γ_{sub} (J/m ²)	Typical use
Si(100)	1.24	Semiconductor devices, MEMS, standard research substrates
Glass (SiO ₂)	0.30	Optical coatings, TCO films, display applications
Sapphire (Al ₂ O ₃)	1.40	LED substrates, GaN epitaxy, high-temperature applications



4.5 Quick Presets

Preset	Configuration	Physics demonstrated
High Rate	Cu DC 480W 6mTorr	High yield, moderate scattering, centre-peaked profile
Uniform	Cu+Al dual gun 12mTorr	Moderate scattering, uniformity optimisation
Reactive	Al RF 20% O2 18mTorr	Berg hysteresis, target poisoning, yield suppression
Co-Sputter	Cu+Al+Zn three guns 10mTorr	Composition tracking, unequal yields at equal power



5 - Running a Simulation

Step 1 - Set parameters: Configure all parameters in left sidebar: target material, operating mode, power, working pressure, target-substrate distance, substrate selection, and gas composition.

Step 2 - Check the Recipe Advisor: Expand the Intelligent Recipe Advisor in the right sidebar. Review the classification and address any flagged issues before proceeding.

Step 3 - Set a thickness target (optional): In Process Controls, enter target film thickness in nm and click Set Target. Simulation pauses automatically when this thickness is reached.

Step 4 - Start the simulation: Click Start Process in the bottom control strip. Particles appear on the chamber canvas within 1-2 seconds. Process Diagnostics panel updates in real time.

Step 5 - Monitor the diagnostics: Watch the Process Diagnostics panel for live values of yield, MFP, transport regime, deposition rate, uniformity NU%, growth mode, and delta-gamma. The time series chart shows deposition rate history.

Step 6 - Pause or stop: Click Pause to pause without resetting. Click Reset Film to clear accumulated thickness and restart tracking.

Step 7 - Export data: Click Export CSV to download simulation data containing time series, radial thickness profile, and process summary.

Tip: For your first run, use the "High Rate" preset - it immediately demonstrates the core physics with a physically meaningful Cu DC deposition recipe.

6 - Reading the Diagnostics Panel

6.1 Process Controls

Shows elapsed process time in hh:mm:ss and current simulation status (RUN/PAUSED/IDLE). Thickness Target subsection allows setting auto-pause thickness in nm.

6.2 Process Diagnostics

All live physics outputs updated every frame.

Diagnostic	Units	Physical meaning
Avg Rate	atoms/s	Time-averaged deposition flux at substrate centre
Max Yield	atoms/ion	Peak sputtering yield across all active guns at current ion energy
Base Pressure	Torr	Fixed reference: 1.0×10^{-6} Torr (turbomolecular-pumped)
Working Pressure	mTorr	Current Ar working gas pressure as set by slider
Transport Ratio (chi)	dimensionless	d/lambda ratio of target-substrate distance to MFP
Transport Regime	-	Ballistic ($\chi < 0.5$) / Moderate ($0.5 \leq \chi \leq 2.0$) / High scattering ($\chi > 2.0$)
Film Uniformity	%	NU% non-uniformity across substrate diameter
Process Status	-	Good / Usable with caution / Not recommended
Growth Mode	-	FM (Frank-van der Merwe) / VW (Volmer-Weber) / SK (Stranski-Krastanov)
Delta-gamma	J/m ²	Bauer free energy criterion. ≤ 0 favours FM; > 0.5 favours VW
Substrate	-	Si(100) / Glass / Sapphire

6.3 Active Recipe

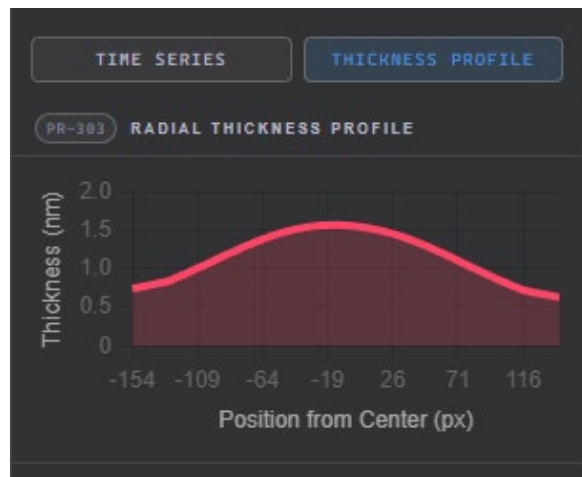
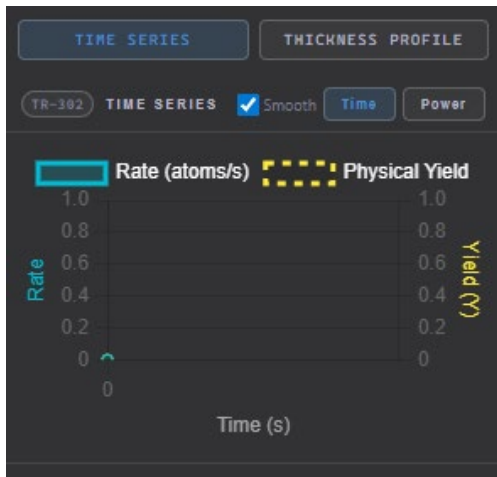
Summarises current gun configuration - materials, modes, power settings, total power, working pressure, and current film composition for co-sputtering. Updates automatically when parameters change.



6 - Reading the Diagnostics Panel

6.4 Time Series and Thickness Profile

- **Time Series:** deposition rate (atoms/s) vs elapsed process time. Confirms steady-state deposition.
- **Thickness Profile:** accumulated film thickness across substrate diameter (radial profile). Flat = good uniformity; peaked = centre-heavy.



7 - Intelligent Process Advisory Module

The module evaluates the complete set of configured parameters against the physics engine and classifies the recipe into one of three categories.

Classification	Criteria	Recommended action
Good combination	All five criteria within validated bounds	Proceed with simulation. Results are physically meaningful.
Usable with caution	One criterion deviates: $\chi \geq 2.0$, NU% < 85%, or one process flag raised	Review flagged parameter. Simulation will run but one aspect may be sub-optimal.
Not recommended	Two or more criteria outside bounds, NU% < 60%, power < 100W, or $\chi \geq 3.2$	Adjust parameters before running. Results may not be physically meaningful.

The five criteria evaluated: (i) scattering ratio $\chi = d/\lambda$; (ii) film non-uniformity NU%; (iii) applied power vs minimum stable discharge; (iv) working pressure vs discharge stability limit (0.5 mTorr); (v) target-to-substrate coverage ratio and reactive gas fraction.

The advisor panel is collapsed by default. Click the header row to expand it.

8 - Co-Sputtering

SputterOne supports simultaneous operation of up to three independent magnetron guns for co-sputtering simulation.

To enable co-sputtering:

1. Toggle on Gun 2 and/or Gun 3 using the switch next to the gun header.
2. Set target material, mode, and power for each active gun independently.
3. Start simulation - chamber canvas shows three distinct plasma confinement regions.
4. Monitor film composition in real time in the Active Recipe section.

Film composition is tracked using yield-weighted superposition. Composition fraction of material k:

$C_k = D_k / \sum(D_i)$, where D_k is cumulative deposited thickness from source k.

Equal power on chemically distinct targets does not produce equal film composition.

Example: Cu (200W DC) + Al (200W DC) at 5 mTorr - Cu yield (~2.1 atoms/ion) is approximately 3x higher than Al yield (~0.65 atoms/ion) at same power, so film will be approximately 75% Cu and 25% Al, not 50:50.

9 - Reactive Sputtering

Reactive sputtering is activated by increasing O2 fraction slider above 0% in Gas Composition section.

Physical behaviour modelled:

- **Target poisoning:** as O2 fraction increases, target surface progressively forms oxide compound layer, reducing effective yield.
- **Hysteresis:** transition from metallic to compound mode and return path occur at different O2 fractions, producing Berg hysteresis loop.
- **Yield suppression:** poisoning factor f_{poison} ranges from 1.0 (fully metallic) to 0.12 (fully poisoned), consistent with experimentally reported compound-to-metallic yield ratios.

Tip: To observe hysteresis, increase O2 fraction from 0% to 30% slowly (2% at a time, pausing between each step). Then decrease back to 0% - the return path shows a different yield vs O2 curve demonstrating hysteresis.

Known limitation: The current model does not account for pumping speed dependence, chamber wall gettering, or compound yields within a partially poisoned target. For quantitative reactive process design, consult dedicated Berg-model solvers.

10 - Exporting Data

Click Export CSV in the bottom control strip. The CSV file contains three sections.

Section	Contents
Time Series	Elapsed time (s), deposition rate (atoms/s), accumulated thickness (nm), yield (atoms/ion), uniformity NU%, transport regime - logged every second
Radial Profile	Binned thickness distribution across substrate diameter. Use for uniformity analysis and comparison between pressure conditions.
Process Summary	All configured parameters and final output metrics at end of run: total thickness, elapsed time, average rate, final NU%, growth mode, delta-gamma.

11 - Tutorial and Laboratory Modules

Access Tutorial and Laboratory Modules from the navigation bar.

14-Chapter Tutorial

Sequential tutorial progressing from vacuum science and plasma fundamentals through sputtering yield, gas-phase transport, reactive sputtering, co-sputtering, and film growth modes. Each chapter includes:

- Physics explanation with governing equations
- Animated SVG diagram illustrating the concept
- Key equation with variable definitions
- A peer-reviewed real-world reference callout

Five Virtual Laboratory Modules

Lab	Title	Key learning objective
Lab 1	Pressure vs Transport Regime	Map working pressure to ballistic/moderate/diffusive regime and observe effect on deposition profile
Lab 2	Yield vs Power and Material	Compare sputtering yield across materials and power levels; verify Yamamura-Bohdansky energy dependence
Lab 3	Geometry Optimisation for Uniformity	Optimise target-substrate distance and target diameter for minimum NU%
Lab 4	Reactive Sputtering Hysteresis	Map O2 fraction vs yield hysteresis loop; identify poisoning onset
Lab 5	Co-Sputtering Composition Control	Demonstrate that equal power on distinct targets gives unequal composition; design a target composition

12 - Physics Reference

12.1 Sputtering Yield Model

SputterOne uses the Yamamura-Bohdansky semi-empirical formulation:

$$Y(E) = Q \times S_n^*(\epsilon) \times [1 - (E_{th}/E)^{2/3}]^{2.5}$$

where Q is material-specific yield parameter, $S_n^*(\epsilon)$ is nuclear stopping cross-section in Lindhard reduced units, and E_{th} is threshold energy. Parameters from Yamamura and Tawara (1996) IPPJ-AM-26.

12.2 Gas-Phase Transport

Mean free path from kinetic theory:

$$\lambda = k_B T / (\sqrt{2} \times \pi \times d_{mix}^2 \times P)$$

where d_{mix} is effective hard-sphere diameter of gas mixture, $T = 300$ K, P is working pressure. Transport regime: Ballistic ($\chi < 0.5$), Moderate ($0.5 \leq \chi \leq 2.0$), High scattering ($\chi > 2.0$).

12.3 Discharge Model

Parametric discharge: $I = P/V$, $E_{ion} = V \times f_{type} \times f_B$.

- DC mode: $f_{type} = 1.0$.
- RF mode: $f_{type} = 0.60$ (self-bias reduction at 13.56 MHz).
- Magnetic confinement factor f_B ranges 0.85-1.10.
- RF self-bias produces bimodal ion energy distribution - SputterOne uses mean energy.
- Model does not resolve plasma balance, electron transport, or pressure-dependent ionisation.

12.4 Reactive Sputtering

$$Y_{eff} = Y_{clean} \times f_{poison} \times f_{hys}.$$

f_{poison} in $[0.12, 1.0]$ decreases monotonically with O₂ fraction. f_{hys} introduces asymmetry between forward and return paths, approximating Berg hysteresis loop without solving oxygen balance or adsorption kinetics.

12.5 Growth Mode Classification

Bauer criterion:

$$\Delta\gamma = \gamma_{film} + \gamma_{int} - \gamma_{sub}.$$

Interfacial energy from geometric mean:

$$\gamma_{int} = \gamma_{film} + \gamma_{sub} - 2\sqrt{\gamma_{film} \times \gamma_{sub}}.$$

Classification: FM ($\Delta\gamma \leq 0$, high mobility); SK ($0 < \Delta\gamma \leq 0.5$ or mismatch $> 7\%$); VW ($\Delta\gamma > 0.5$ J/m²). Surface free energy values from Ohring (2002).

13 - Known Limitations

Limitation	Description
Pressure-dependent rate trend	The simulator may predict higher deposition rate at lower pressure while experiment shows the opposite. Use for transport regime classification, not absolute rate comparisons across pressures.
RF/DC ordering	SputterOne predicts DC > RF deposition rate. Experimental systems at 1-3 mTorr may show RF > DC due to capacitive flux enhancement not modelled. Difference is small (2-3%).
No plasma kinetics	Discharge voltage and ion current are parametric estimates. Pressure-dependent ionisation efficiency, sheath dynamics, and electron temperature are not resolved.
Deposition rate units	Display yield and deposition rate are calibrated relative values, not absolute SI-traceable quantities. Do not use for absolute process recipe design.
Heavy metal yield accuracy	W, Ta, Mo show 43-70% deviation from SRIM due to simplified LSS stopping power vs ZBL potential. Material ordering is correct but absolute values should be treated with caution for refractory metals.
Atomistic surface processes	Adatom diffusion, surface roughness, redeposition, and temperature-dependent hopping rates are not modelled. Growth mode is thermodynamic classification, not kinetic simulation.
macOS desktop app	The desktop application currently supports Windows only. macOS support is under development.

14 - How to Cite

Journal paper citation

Gururajan, B. (2026). *SputterOne: A Physics-Informed Interactive Magnetron Sputtering Simulator Enabling a Process Digital Twin*. Advanced Theory and Simulations (Wiley). Manuscript ID 4581533. DOI: [to be assigned upon publication].

Software repository citation

Gururajan, B. (2026). *SputterOne v1.0.1* [Software].
<https://balaji-gururajan90.github.io/Optoelectronics-device-laboratory-sputterone.html>

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